BENJAMIN AGALISI

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PROFESSIONAL SUMMARY

Analytical Level 100 Computer Engineering student at Ghana Communication Technology University (GCTU) with core competence in **C++**, low-level **C programming**, **Arduino embedded systems**, and database security. Proven ability to bridge hardware and software by interfacing microcontrollers with sensors, designing dynamic backend systems (PHP/MySQL), and leading technical engineering teams. Seeking a Computer Engineering Internship to apply hands-on embedded programming and software design skills in an industry environment.

EDUCATION

Ghana Communication Technology University (GCTU) Accra, Ghana

Bachelor of Science in Computer Engineering (Level 100) | 2025 – Present

- **Relevant Coursework:** C/C++ Programming, Circuit Theory (Thevenin/Norton Theorems), Differential Equations, Mechanics & Electromagnetism, Database Systems.
- **Academic Leadership:** Appointed Engineering Team Lead; directed repository workflows (Git/GitHub), code reviews, and technical demonstrations.

Presbyterian Senior High School, Adeiso : Eastern Region, Ghana

West African Senior School Certificate Examination (WASSCE)

- **Leadership:** Secretary of the Science Club; organized regional STEM competitions, exhibitions, and peer study groups.

Abokobi Presby Basic School — Greater Accra Region, Ghana

Basic Education Certificate Examination (BECE)

- **Leadership:** School Prefect; led student executive council and coordinated campus assemblies.

TECHNICAL SKILLS

- **Embedded Hardware & Systems:** Arduino Microcontrollers, C++ Embedded Programming, Breadboard Circuit Design, Sensor & Actuator Integration, PlatformIO (VS Code).

- **Programming Languages:** C++, C, PHP, SQL, HTML5, CSS3, JavaScript (Basic).
- **Databases & Backend:** MySQL, PHP Data Objects (PDO), Prepared Statements, Role-Based Access Control (RBAC), XAMPP Environment.
- **Cybersecurity & Tools:** Secure Coding Practices, SQL Injection Vulnerability Analysis, Hack The Box, Git & GitHub, Visual Studio Code, Linux CLI.
- **Data & Analytics:** Microsoft Excel (Advanced Formulas: `COUNTIF`, `TEXT`, Dynamic Sorting), Tableau Visualization.
- **Languages:** English (Professional Fluency), French (Intermediate Practice), Buli (Native).

ENGINEERING & HARDWARE PROJECTS

Arduino Embedded Control & Sensor System | *C++, Arduino IDE / PlatformIO*

- Programmed microcontroller control logic in C++ to process real-time input from distance sensors and physical switches.
- Built hardware circuit prototypes on breadboards incorporating LED timing sequences and automated sensor-driven triggers.

AGALISI — C/C++ Text & Frequency Analyzer | *C/C++, Data Structures, Git*

- Spearheaded a 6-member team to architect a low-latency console application for text corpus analysis.
- Implemented dynamic memory allocation, pointer arithmetic, string parsing, and the standard `qsort` algorithm for dynamic frequency evaluation.

Web Management & Security Portal | *PHP, MySQL, HTML/CSS, PDO*

- Built a multi-tier database application featuring Role-Based Access Control (RBAC) dashboards (Manager, Receptionist, Admin).
- Enforced backend security standards using PHP PDO prepared statements to safeguard authentication systems against SQL injection attacks.

Earth Guard AI & Flood Warning Architecture | *IoT & Edge Computing Concepts*

- Designed system proposals integrating satellite observation with localized microcontroller sensor nodes for environmental monitoring and flood risk detection.

LEADERSHIP & COMMUNITY ENGAGEMENT

- **STEM Tutoring & Curriculum Design:** Authored structured lesson plans and delivered targeted instruction in Mathematics and ICT for Junior High School students.
- **Youth Leadership & Advocacy:** Active participant in youth development initiatives and civic outreach programs across Ghana